# Lab Experiment-10

**Aim**

To implement a transformation-based Part-of-Speech (POS) tagging system in Python by assigning initial tags to words and applying a set of transformation rules to improve the tagging accuracy.

**Algorithm**

1. Start the program.
2. Read a sentence from the user.
3. Split the sentence into individual words using regular expressions.
4. Assign an initial tag (NN) to words and PUNCT to punctuation marks.
5. Apply transformation rules to identify determiners, pronouns, verbs, adjectives, prepositions, and conjunctions.
6. Apply contextual rules to identify verbs ending in -ing when preceded by an auxiliary verb.
7. Display each word with its final POS tag.
8. Stop the program.

**Code**

```
import re

def transformation_based_tagging(sentence):

    words = re.findall(r"\b\w+\b|[.,!?;:]", sentence)

    tagged_words = [

        [word, "NN" if word.isalnum() else "PUNCT"]

        for word in words
```

```
]

determiners = ["the", "a", "an", "this", "that"]

pronouns = ["i", "you", "he", "she", "it", "we", "they"]

auxiliary_verbs = ["is", "am", "are", "was", "were"]

conjunctions = ["and", "or", "but"]

prepositions = ["in", "on", "at", "with", "under"]

adjectives = ["good", "beautiful", "happy", "quick"]

verb_tags = {

    "reads": "VBZ",

    "writes": "VBZ",

    "plays": "VBZ",

    "read": "VB",

    "write": "VB",

    "play": "VB"

}

for item in tagged_words:

    word = item[0].lower()

    if word in determiners:

        item[1] = "DT"

    elif word in pronouns:
```

```
            item[1] = "PRP"
        elif word in auxiliary_verbs:
            item[1] = "VBD" if word in ["was", "were"] else "VB"
        elif word in conjunctions:
            item[1] = "CC"
        elif word in prepositions:
            item[1] = "IN"
        elif word in adjectives:
            item[1] = "JJ"
        elif word in verb_tags:
            item[1] = verb_tags[word]
    for i in range(1, len(tagged_words)):
        current_word = tagged_words[i][0].lower()
        previous_word = tagged_words[i - 1][0].lower()
        if (previous_word in auxiliary_verbs
                and current_word.endswith("ing")):
            tagged_words[i][1] = "VBG"
    return tagged_words
sentence = input("Enter a sentence: ")
result = transformation_based_tagging(sentence)
```

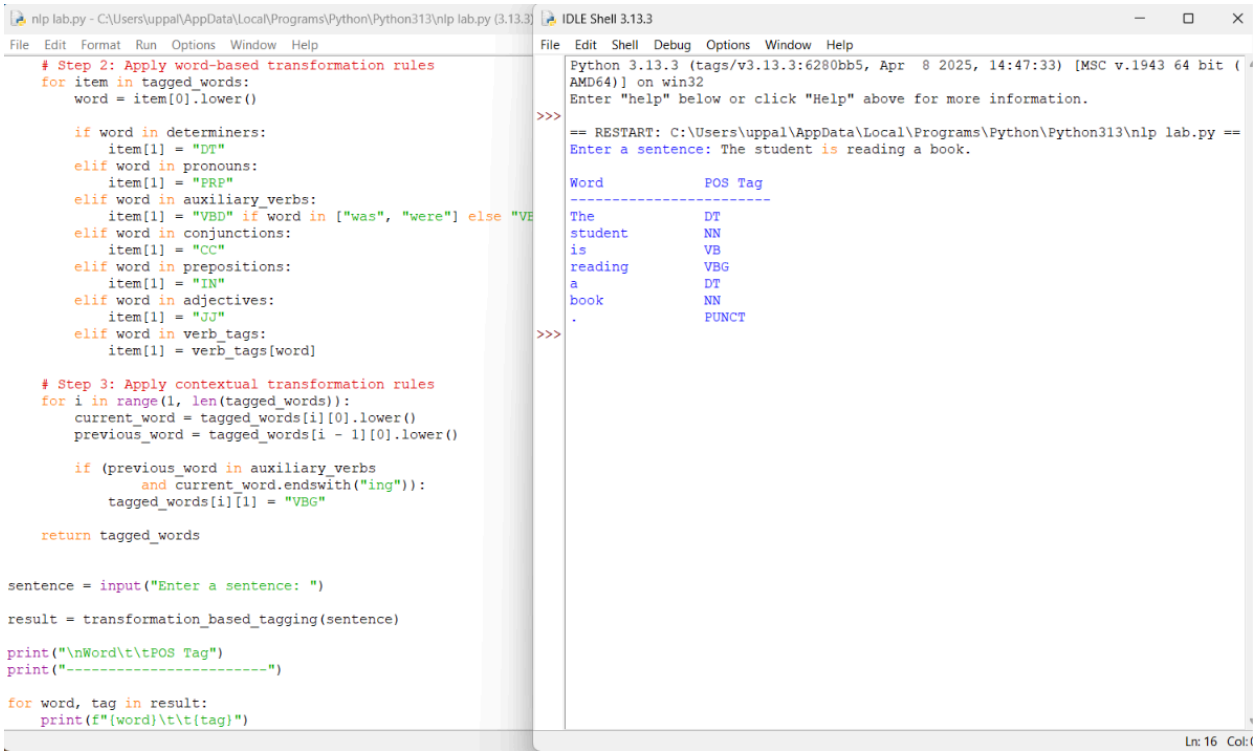
```
print("\nWord\t\tPOS Tag")

print("------------------------")

for word, tag in result:

    print(f"{word}\t\t{tag}")
```

**Output**




**Result**

Thus, a transformation-based POS tagging system was implemented successfully using Python. The program initially assigns default tags and then applies word-based and contextual transformation rules to improve the tags. For example, the word reading is tagged as VBG because it ends in -ing and follows the auxiliary verb is.